Práctica dos: Modelado y control del sistema cardiovascular mediante el modelo Windkessel

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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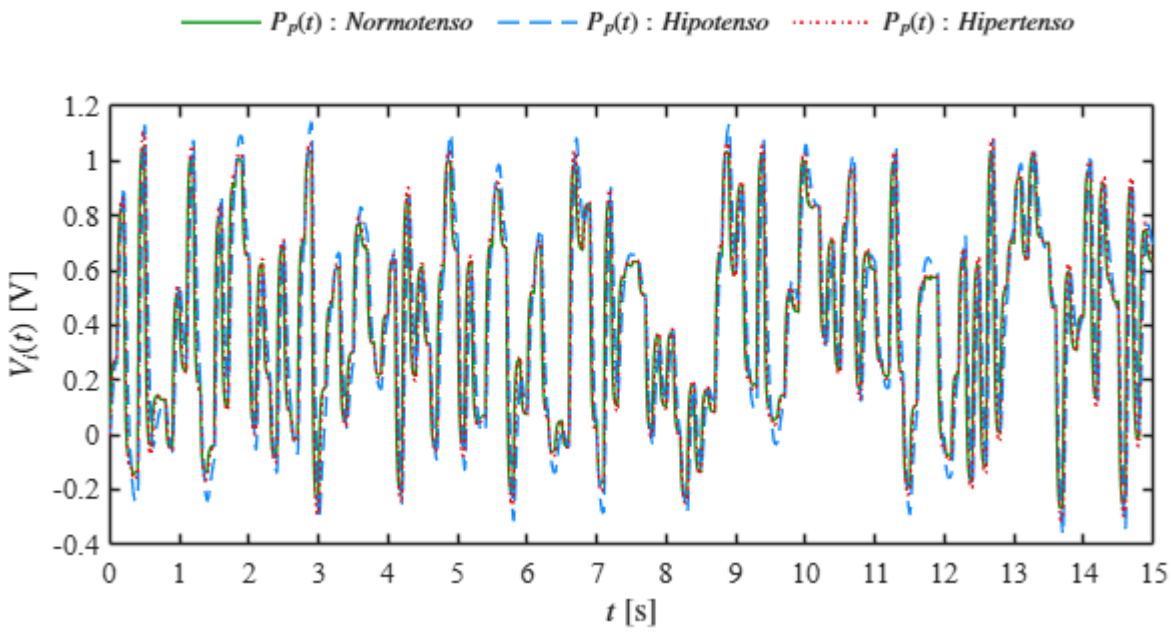
Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('mijo')
parameters.StopTime='15';
```

```
parameters.Solver='ode15s';
parameters.MaxStep='1e-3';
x=sim('mijo',parameters);
t_data=(0:0.1:15)';
t=(0:0.001:15)';
Pa=interp1(t_data,x.Pa,t,'previous');
writematrix(Pa,'signal.xlsx');
```

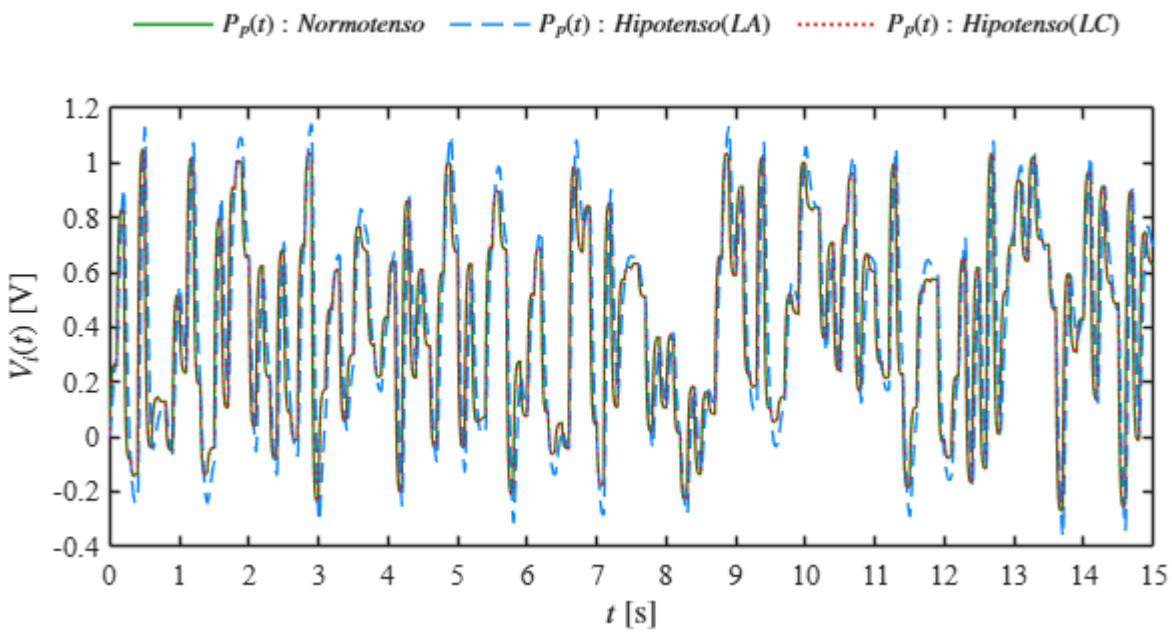
Respuesta en lazo abierto

```
plotSignals(x.t,x.PpN,x.PpL,x.PpH,x.PIDL,x.PIDH,'Individual')
```



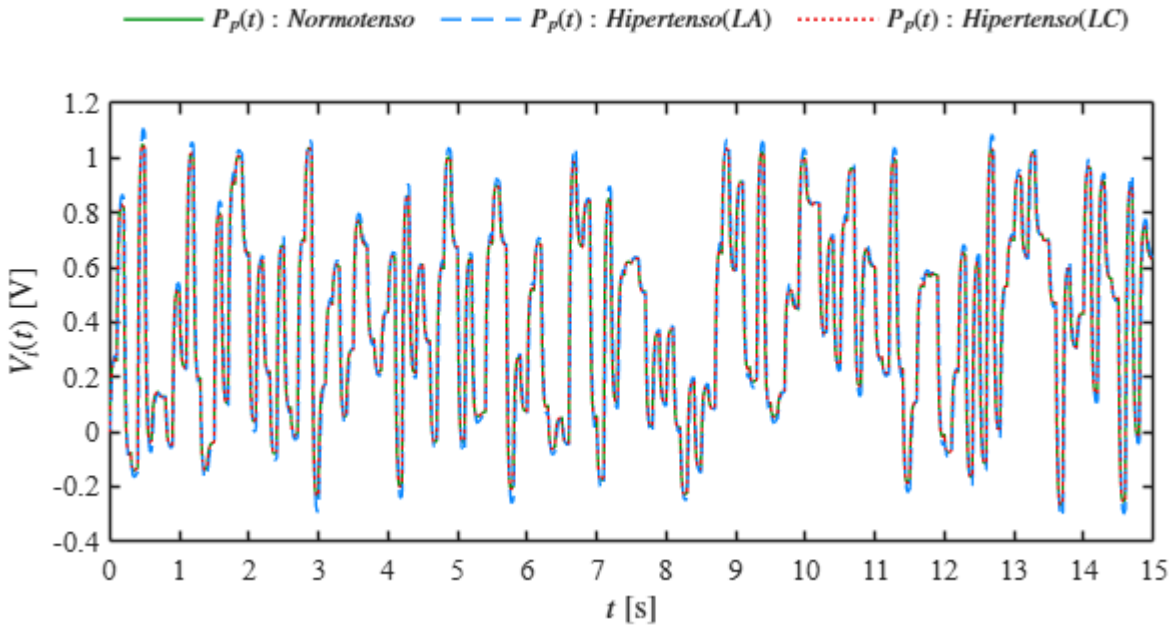
Hipotenso

```
plotSignals(x.t,x.PpN,x.PpL,x.PpH,x.PIDL,x.PIDH,'Hipotenso')
```



Hipertenso

```
plotSignals(x.t,x.PpN,x.PpL,x.PpH,x.PIDL,x.PIDH,'Hipertenso')
```



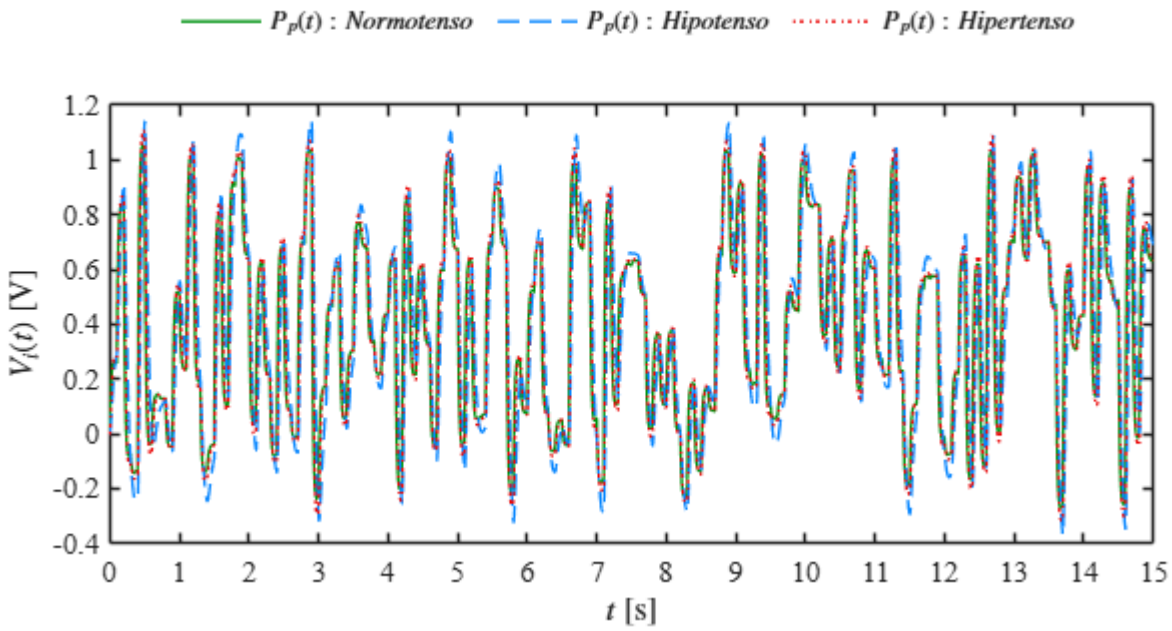
Modelo de EDOs

```
par_hipo=[0.010,0.001,1.000,3.500]; %Z,L,R,C
par_normo=[0.010,0.001,1.000,1.500]; %Z,L,R,C
par_hiper=[0.010,0.001,1.000,1.000]; %Z,L,R,C

PpN=cardio(par_normo,Pa,t);
PpL=cardio(par_hipo,PpN,t);
```

```
PpH=cardio(par_hiper,PpN,t);

plotSignals(t,PpN,PpL,PpH,x.PIDL,x.PIDH,'Individual')
```




```
function Pp=cardio(par,Pa,t)
    Z=par(1);L=par(2);R=par(3);C=par(4);dt=t(2);
    n=round(t(end)/dt);


    FL=zeros(length(t),1); FL(1)=0;
    Pp=zeros(length(t),1); Pp(1)=0;

    for i=1:n
        FL(i+1)=FL(i)+(-(1/L)*Pp(i)+(1/L)*Pa(i))*dt;
        Pp(i+1)=Pp(i)+((1/C)*FL(i)-(1/(Z*C)+1/(R*C))*Pp(i)+1/(Z*C)*Pa(i))*dt;
    end


end
```

Función: Respuesta de la simulación

```
function plotSignals(t,PpN,PpL,PpH,PIDL,PIDH,signal)
    set(figure(),'color','white')
    set(gcf,'units','inches','position',[1,1,7,3.5])
    set(gca,'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;

    clr1=[0, 148, 18]/255;
```

```
    clr2=[8, 135, 255]/255;
    clr3=[235, 9, 9]/255;
    order=[clr1;clr2;clr3]; colororder(order)



    if signal=="Individual"

plot(t,PpN,'-',t,PpL,'--',t,PpH,':','LineWidth',1,'MarkerSize',5,'MarkerIndic
es',...
            1:1e3:length(t));
        L=legend('$P_{p}(t): Normotenso$','$P_p(t): Hipotenso$','$P_p(t):
Hipertenso$');

set(L,'interpreter','latex','location','northoutside','orientation','horizont
al','Box','off')

        xlabel('$t$ [s]','interpreter','latex')
        ylabel('$V_i(t)$ [V]','interpreter','latex')
        xlim([0 15]); xticks(0:1:15)
        ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
    elseif signal=="Hipotenso"

plot(t,PpN,'-',t,PpL,'--',t,PIDL,':','LineWidth',1,'MarkerSize',5,'MarkerIndi
ces',...
            1:1e3:length(t));
        L=legend('$P_{p}(t): Normotenso$','$P_p(t): Hipotenso (LA)
$','$P_p(t): Hipotenso (LC)$');

set(L,'interpreter','latex','location','northoutside','orientation','horizont
al','Box','off')

        xlabel('$t$ [s]','interpreter','latex')
        ylabel('$V_i(t)$ [V]','interpreter','latex')
        xlim([0 15]); xticks(0:1:15)
        ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
    elseif signal=="Hipertenso"

plot(t,PpN,'-',t,PpH,'--',t,PIDH,':','LineWidth',1,'MarkerSize',5,'MarkerIndi
ces',...
            1:1e3:length(t));
        L=legend('$P_{p}(t): Normotenso$','$P_p(t): Hipertenso (LA)
$','$P_p(t): Hipertenso (LC)$');

set(L,'interpreter','latex','location','northoutside','orientation','horizont
al','Box','off')

        xlabel('$t$ [s]','interpreter','latex')
        ylabel('$V_i(t)$ [V]','interpreter','latex')
        xlim([0 15]); xticks(0:1:15)
```

```
        ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
    end

    exportgraphics(gcf,[signal,'_matlab.pdf'],'ContentType','vector')



end
```